

ECON 101: Macroeconomics

Quiz 2

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Instructions

Answer all questions on the answer sheet. Show all work for numerical problems. Multiple-choice questions have one correct answer.

Part A – Multiple Choice (1 mark each)

1. Suppose beef and chicken are substitutes. If the price of beef rises, holding all else constant:

- a) Demand for chicken decreases.
- b) Supply of chicken decreases.
- c) Demand for chicken increases.
- d) Quantity demanded of chicken decreases.

2. Which of the following transactions is **included** in Canada's GDP?

- a) A Canadian student buys used textbooks from another student.
- b) A Canadian firm sells lumber to a U.S. company.
- c) A Canadian tourist purchases a meal in Paris.
- d) A Canadian household does its own home repairs.

3. Real GDP differs from nominal GDP because real GDP:

- a) Includes only goods, not services.
- b) Is adjusted for population size.
- c) Values output at constant prices.
- d) Excludes investment spending.

4. If the GDP deflator increases from 120 to 132, the inflation rate is approximately:
- a) 10%
 - b) 12%
 - c) 9%
 - d) 8%
5. Which of the following would **not** increase GDP?
- a) Higher consumption of new cars.
 - b) More exports of manufactured goods.
 - c) An increase in unpaid household work.
 - d) Greater investment in factories.

Part B – Numerical Problems (15 marks total)

1. Market Equilibrium (4 marks) The market demand and supply for pizzas in Toronto are given by:

$$Q_D = 200 - 5P, \quad Q_S = 20 + 3P,$$

where Q is the number of pizzas and P is the price in dollars.

- a) Find the equilibrium price and quantity.
- b) If the government imposes a price floor at \$25, calculate the resulting surplus or shortage.

2. Expenditure Approach to GDP (4 marks) Suppose the following Canadian macroeconomic data (in billions):

- Consumption: \$1,200
- Investment: \$500
- Government spending: \$600
- Exports: \$450
- Imports: \$550

- a) Calculate nominal GDP using the expenditure approach.
- b) If population = 37 million, compute GDP per capita.
- c) If potential GDP = \$3,000 billion, calculate the output gap as a percentage.

3. Real vs. Nominal GDP (7 marks) An economy produces two goods: milk and computers.

Year	Quantity Milk	Price Milk	Quantity Computers	Price Computers
2023	1,000	\$2	50	\$500
2024	1,200	\$3	55	\$550

- a) Compute nominal GDP for 2023 and 2024.
- b) Compute real GDP for 2024 using 2023 as the base year.
- c) Calculate the GDP deflator for 2024.
- d) What is the inflation rate between 2023 and 2024?